METRIC

MIL-P-63252A (AR)
AMENDMENT 2
14 May 1998
SUPERSEDING
MIL-P-63252A (AR)
AMENDMENT 1
10 October 1996

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, M795
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-P-63252A (AR), dated 28 September 1995 and is approved for use by the US Army Armament, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

Entire Specification MIL-P-63252

Delete the word “penetrameter(s)” and replace with “Image Quality Indicator(s) (IQI’s)”

PAGE 1

* 2.1 Government documents.

Delete “Government documents” and replace with “Documents”

* 2.1.1 Specifications, standards, and handbooks.

Delete in its entirety and replace with the following:

AMSC N/A FSC 1320
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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"2.1.1 Non government publications. The following documents form a part of this document to the extent specified herein.

American Society for Testing and Materials

E1742 - Standard Practice for Radiographic Examination.

(Copies may be purchased directly from the American Society for Testing and Materials). American Society For Nondestructive testing Inc, 1711 Arllingate Ln, P.O. Box 28518, Columbus, OH 43228-0518.

American National Standards Institute

CP 189 - Standard for Qualification and Certification of Nondestructive Testing Personnel.

(Copies may be purchased directly from the American Society for Nondestructive Testing Inc.) American Society For Nondestructive Testing Inc, 1711 Arlingate Ln, P.O. Box 28518, Columbus, Oh 43228-0518."

PAGE 2

* 2.1.2 Other Government documents, drawings and publications.

Add: "10542907 - Obturator"

PAGE 6

3.4.1 Mating surfaces. Change:

From:

"The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile."

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To:

"The mating surfaces of threads shall be free of caked explosive or exudate and all threads shall be free of foreign material. If a chemical cleaner is required it shall be approved by the procuring agency. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile."

PAGE 7

3.7 Exudation: Delete in its entirety.

* 3.8.2 Protection of the rotating band.

Add:

"3.8.2.1 Rotating band. The surface of the rotating band lands shall be examined for surface defects, damage and contamination.

a. On the forward land, there shall be no evidence of raised copper due to nicks, digs or damage of any kind and there shall be no paint or foreign matter on the outer diameter.

b. On the aft three lands, any nicks or digs extending the 1/2 the land width or less are acceptable, provided the maximum band diameter is not exceeded.

c. The surface of the overlay band shall have no cavities that exceed .238mm (3/32 inch) in the greatest dimension.

d. On the surface of the overlay band the number of cavities equal to or less than .238mm (3/32 inch) in the greatest dimension shall not exceed one (1) per 45 degree segment of the band. However one (1) 6.45 square mm (1 square inch) area per band is allowed a maximum of four cavities which exceeds .238mm (3/32 inch) in the largest dimension.

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e. On the surface of the overlay band one tightly closed seam in the area of the weld start stop zone is allowed.

f. Disbond (pick out) as indicated by a ragged appearance forward or aft of the rotating band is allowed.

g. The overlay rotating band and band seat areas shall have no other discontinuities, cracks, lack of bonding, lack of fill or other metal defects other than those herein defined.

h. Severity of defects shall be in accordance with visual standards.

PAGE 15

4.4.2.2 Projectile and Liner.

Special

a. Change:

From:

"Presence of explosive on exterior of shell body, threads and forward face."

To:

"Presence of explosive or exudate on exterior of shall body, threads and forward face."

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4.4.2.3 Projectile, prior to assembly of supplementary charge.

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Major

104 Swage missing or incomplete

Delete: “Gage” and Substitute: “Visual”

105 Inside diameter of liner after swaging

Delete: “Visual” and Substitute: “Gage”

106 True position of inside diameter of liner, W/Pitch diameter of thread

Delete: “W/Pitch diameter of thread” and Substitute: “W/Minor diameter of thread”

4.4.2.5 Projectile, prior to assembly of supplementary charge.

Delete Minor 202 “Liner removable by hand” in its entirety.

PAGE 18

4.4.2.4 Projectile Assembly.

Minor

201 Grommet missing, damaged or improperly assesmbled.

Change From:

“Grommet missing, damaged or improperly assembled.”

To:

“Rotating band cover, damaged or improperly placed.”

PAGE 20

Add new paragraph 4.4.2.8 as included page 20a.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Obturator	1 OF 1	10542907
			NEXT HIGHER ASSEMBLY 9312769

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of rear I.D. with forward I.D.	Level III	3.2	SMTE
102	Diameter outside, front	Level III	3.2	SMTE
103	Diameter outside, rear	Level III	3.2	SMTE
104	Diameter inside, front	Level III	3.2	SMTE
105	Depth of forward inside diameter	Level III	3.2	SMTE
106	Thickness of wall at rear	Level III	3.2	SMTE
107	Length, total	Level III	3.2	SMTE
108	Evidence of poor workmanship	Level III	3.8	Visual

Notes:

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* 4.4.3.1.1 Base separation.

Delete in its entirety and replace with:

"4.4.3.1.1 Base separation. Projectiles showing the largest base separation as determined by radiographic examination prior to post cycle heat treatment shall be included in the post cycle heat treatment, placed in the coolest area of the igloo and retained for sectioning. A minimum of eight (8) samples shall be used for the first article and the first three (3) post cycle heat treatment lots have been accepted, samples shall be selected at the rate of one (1) out of seven hundred (700) projectiles of fraction thereof with the minimum sample size of tow (2) samples per post cycle heat treatment inspection lot. When a production lot has been rejected for base separation the sample size shall revert back to the minimum eight (8) samples per post cycle heat treatment lot. If insufficient number of samples are available with base separation based on radiographs, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. Any projectile after sectioning which has base separation in excess of that permitted by the applicable requirement, shall be classed defective, and the entire post cycle heat treatment lot subjected to recycle. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4. Any lot which fails after being post cycle heat treated three times shall be rejected."

PAGE 23

4.4.3.3 Exudation. Delete in its entirety.

* 4.5.1.1 Determination of projectile cavitation.

Delete "MIL-STD-453" and replace with "ASTM E 1742"

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* 4.5.1.2 Check of sensitivity of radiographic technique.

Delete “MIL-STD-453” and replace with “ASTM E 1742” two places.

PAGE 26

Figure 2 top right note.

Delete “MIL-STD-453” and replace with “ASTM E 1742”

PAGE 27

4.5.3 Exudation. Delete in its entirety.

4.5.1.6 Operator qualification.

Delete “MIL-STD-410” and replace with “ANST CP 189” two places.

* 4.5.2 Proving ground test.

Delete “The sample shall be assembled with a dummy or inert loaded fuze.” And replace with “The sample shall be assembled with a PD fuze set to SQ.”

Delete “Camera coverage shall be provided to observe the projectile in flight in order to record any malfunctions (see 6.12.” and replace with “Camera coverage shall be provided to observe the projectile during initial flight to record any malfunctions.”

* 4.3.2.1 First article and production samples.

Delete “-54” and replace with “-51” two places.

6.8.1 Radiographic film sources. Delete in its entirety and substitute the following:

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"6.8.1 Radiographic film sources. Approved list of film types and vendors for use in radiographing this projectile:

Eastern Kodak Company
1187 Ridge Road, West
Rochester, NY 14650

Film Type: AA, T, AX-1, MX

Agfa Corporation
NDT Systems
100 Challenger Road
Ridgefield Park, NJ 07660

Film Type: Agfa D7

Fuji Industrial Radiographic Products
Corporate Headquarters
200 North Avenue
Larchmont, NY 10538

Film Type: Fuji 100"

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian
Army-AR

Preparing activity:
Army-AR

(Project-1320-0009)

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